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A chemical kinetic study of the oxidation of dibutyl-ether in a jet-stirred reactor

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ABSTRACT

The oxidation of dibutyl-ether, a potential lignocellulosic biofuel was studied in a jet-stirred reactor. Fuel-lean, stoichiometric and fuel-rich mixtures were oxidized at a constant fuel mole fraction (1000 ppm), at temperatures ranging from 470 to 1250 K, pressures of 1 and 10 atm, and constant residence time (70 and 700 ms, respectively). The mole fraction profiles obtained through sonic probe sampling, gas chromatography and Fourier transform infrared spectrometry were used to develop a detailed kinetic mechanism for the oxidation of DBE. The carbon neighboring the ether group was found to be the most favorable site for H-abstraction reactions and the chemistry of the corresponding fuel radical drives the overall reactivity. The fuel concentration profiles indicated strong low-temperature chemistry at both pressures: two negative temperature coefficient regions were observed, notably on the 10 atm experiments. This unusual behavior was investigated by means of a rate of production analysis using the mechanism developed in this work. This analysis showed that the low-temperature reactivity stems from the low-temperature chemistry of the fuel and of smaller species such as C_4H_9 which are triggered at different threshold temperatures. The proposed mechanism shows good performances for representing the present experimental data, as well as ignition delay time and flame speed data available from the literature.

1. Introduction

Recent emission regulations as well as growing concerns about environmental issues highlighted the need for more efficiency, sustainability and emission control in the transportation sector. A way to tackle these considerations is the use of biofuels, which have been widely studied over the recent years. Alcohols, that are mainly produced by sugar fermentation, are considered first generation biofuels which can compete with food production; this is a major drawback. Therefore, a search for alternative feedstocks for the next generation biofuels is growing.

Lignocellulose is more abundant than sugar in biomass [1]. It can be converted to furfural which is proved to be an interesting platform for the synthesis of a wide range of useful molecules including fuel candidates such as cyclopentanone or pentanol [2]. Decarbonylation of furfural leads to furan, which can be hydrogenated to produce tetrahydrofuran (THF). Butanol is then produced from the THF ring opening [3] and a further dehydration reaction leads to the production of dibutyl-ether [4] (DBE) which is an interesting alternative fuel for diesel engines. DBE is liquid in normal condition, and is not miscible with water. This is an advantage in terms of preventing corrosion issues in engines. Its high cetane number (~100) [5] and its energy content, which is comparable to regular diesel fuel, make it a better alternative than lighter ethers such as dimethyl ether.

Several studies on DBE combustion in engines were reported in the literature. Beatrice et al. [6] observed a much lower sooting tendency with DBE than with tetradecane in a Diesel engine, and attributed it to its oxygenated structure. It was also underlined that the short ignition delay time observed enabled a better combustion and a lower production of unburned hydrocarbons. Also, Beeckmann et al. [7] studied the spray characteristics of pure DBE and of diesel fuel/DBE blends showing that DBE addition improved atomization and thus led to better performances. Few studies concerning the oxidation kinetics of dibutyl-ether are also available. Mellouki et al. [8] studied the reaction of the OH radical with DBE in a pulsed laser photolysis-laser induced fluorescence apparatus in the temperature range of 230 to 372 K and proposed a rate constant for the reaction DBE + OH → Products. Cai et al. [9] developed

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a detailed mechanism for DBE oxidation including both high- and low-temperature chemistry. It was validated over ignition delay times obtained at low temperature in a laminar flow reactor and laminar flame speed measured using a stagnation flow configuration. More recently, Al Rashidi et al. [10] performed *ab initio* calculations to investigate the isomerization and β-scission reactions of the fuel radicals as well as the H-abstraction reactions from the fuel by H atoms. Calculations were carried out using CBS-QB3 and G4 composite methods, and conventional Transition State Theory (TST) was finally applied to calculate the rate constants of the different reactions considered. Finally, Guan et al. [11] studied high-temperature ignition delay times of DBE/O_2/Ar mixtures in a shock tube for pressures up to 4 atm, for several dilution and equivalence ratios. DBE was found to be more reactive than dimethyl-ether and diethyl-ether, and the mechanism proposed by Cai et al. [9] was used to simulate the experimental results with a fairly good agreement.

In the present study, new kinetic information on DBE oxidation using complementary approaches is provided. Experiments were carried out in a jet-stirred reactor over a large range of conditions to provide insights into the intermediate compounds formed during DBE oxidation. Pressure-dependent rate constants for important reactions of the fuel and its radicals were calculated using the G3B3 compound method. A kinetic mechanism was finally developed and validated using these data and ignition delay times as well as laminar burning velocities data available in the literature [9–11].

2. Experimental apparatus

Experiments were carried out in a fused silica jet-stirred reactor settled inside a stainless-steel pressure resistant jacket. An electrical oven enabled to perform experiments up to c.a. 1280 K. The temperature within the reactor was continuously monitored by a Pt/Pt-Rh thermocouple located inside a thin wall fused silica tube to prevent catalytic reactions on the metallic wires. Fuel mole fraction was 1000 ppm for all experiments, pressure and residence time (τ) were held constant at 1/10 atm and 0.07 s (at 1 atm) and 0.7 s (at 10 atm). The reactive mixtures were highly diluted by nitrogen to avoid high heat release and experiments were performed at temperatures ranging from 470 to 1280 K. The liquid fuel was atomized by a nitrogen flow and vaporized through a heated chamber. Reactants were brought separately to the reactor to avoid premature reactions and then injected by 4 injectors providing stirring. Flow rates of the diluent and reactants were controlled by mass flowmeters. A low-pressure sonic probe was used to freeze the reactions and take samples of the reacting mixtures.

Online analyses were performed after sending the samples via a heated line to a Fourier transform infrared (FTIR) spectrometer for the quantification of H_2O, CO, CO_2, and CH_2O. Samples were also stored at ca. 40 mbar in Pyrex bulbs for further analyses using gas chromatography (GC). Two gas chromatographs with a flame ionization detector (FID) were used: one equipped with a DB624 column to quantify oxygenated compounds and the other one with a CP-Al_2O_3/KCl column to quantify hydrocarbons. Identification of the products was done by GC/MS on a Shimadzu GC2010 Plus, with electron impact (70 eV) as the ionization mode. Oxygen and hydrogen were measured using a GC-TCD equipped with a CP-CarboPLOT P7 column. Thanks to this analytical equipment, species measured were dibutyl-ether (DBE), O_2, H_2O, CO, CO_2, C_2H_4, CH_4, CH_2O, butanal, H_2, methanol, 1-butene, acrolein, propene, ethenyloxybutane, butyl formate, butanol, propanal, ethanol, acetaldehyde, acetone, 2,3-dihydrofuran, and formic acid. The carbon balance was checked for each sample and found to be 100 ± 10%. Uncertainties on temperature measurements and residence time were estimated to be 10 K and less than 5%, respectively. Inlet uncertainties were less

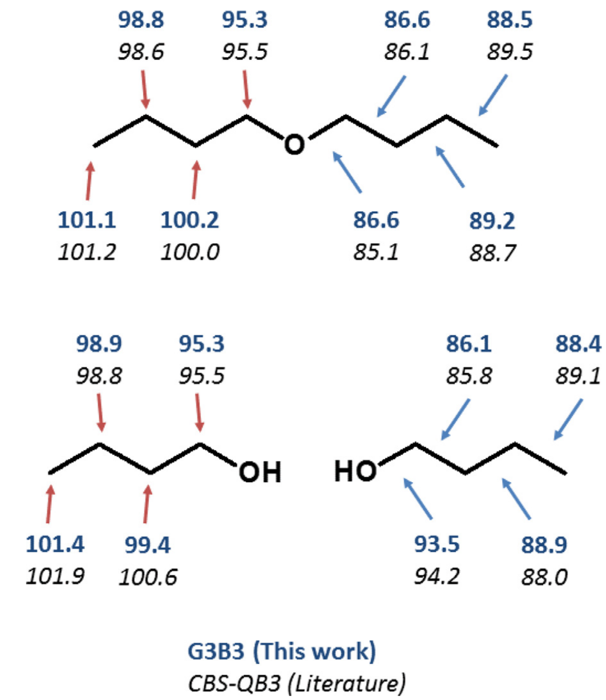

Fig. 1. Bond dissociation energies of DBE and butanol computed with the G3B3 (this work) and with the CBS-QB3 (Cai et al. [9] for DBE, Sarathy et al. [19] and Black et al. [20] for butanol) compound methods in kcal mol^{-1}. Red arrows: C–H bonds, blue arrows: C–C and C–O bonds. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

than 5% for the reactants, and an uncertainty of less than 10% was determined for measured species. The final uncertainty on the different mole fraction profiles depends on all these factors and is estimated to be 15%.

3. Calculations

All theoretical calculations were carried out using the Gaussian 09 program suite [12]. The G3B3 composite method [13] which was applied throughout this study, consists of geometry optimization and vibrational frequency calculations at the B3LYP/6-31g(d) level of theory followed by four single energy calculations at higher levels of theory (QCISD(T)/6-31G(d),MP4/6-31 +g(d),MP4/6-31g(2df,p) and MP2/G3Large). This procedure which is implemented in the Gaussian 09 code through the “G3B3” keyword also includes a scaling factor of 0.96 for the vibrational frequencies and high level corrections based on the G2/97 test set of experimental energies. More details regarding this compound method can be found in the paper of Baboul et al. [13].

The different transition states (TS) geometries of this study were obtained through the “TS” keyword of Gaussian 09 and verified by means of intrinsic reaction coordinate (IRC) calculations. Rate constants for isomerization and β-scission reactions of the primary fuel radicals were then computed using Rice–Ramsperger–Kassel–Marcus (RRKM) theory as implemented in the MESMER software [14] for temperatures ranging between 500 and 2000 K. The Lennard–Jones parameters used for dibutyl-ether are $\sigma = 5.50$ Å and $\varepsilon/k_B = 670.40$ K as calculated using the Tee–Gotoh–Steward correlation model [15] and thermophysical data taken from the literature [16]. Parameters for N_2 which was chosen as the bath gas are $\sigma = 3.74$ Å and $\varepsilon/k_B = 82$ K [17]. The bond dissociation energies of DBE were also computed following the methodology described by Blanksby et al. [18], and compared to those of butanol (Fig. 1). Our results are in a very close agreement with the calculations of Cai et al. [9] (DBE) and Sarathy et al. [19] and

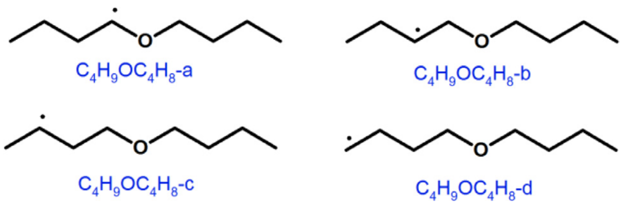

Fig. 2. Nomenclature used in this work for the fuel radicals.

Black et al. [20] (butanol) with the CBS-QB3 compound method. These computations highlight remarkable similarities between the two structures which could justify analogies between these two compounds for estimating the rate constants of H-abstraction reactions.

4. Kinetic modeling

Simulations were carried out with the Chemkin II package. The Perfectly Stirred Reactor (PSR) code [21] was used to perform simulations for the JSR experiments while the SENKIN [22] and PREMIX [23] codes were used to model experimental results available in the literature (ignition delay times in shock tubes and laminar burning velocities respectively). The mechanism used is based on an in house C_0–C_4 core model that was developed by Fenard et al. [24]. This model includes detailed chemistry for various compounds including alcohols, aldehydes and hydrocarbons. The DBE sub-mechanism proposed here is based on a recent work from Cai et al. [9] which originally included 95 species and 341 reactions. Modifications made to this sub-mechanism are based on analogies with butanol and computations, as described in the next two sections. Thermodynamic data for the fuel and its radicals are results from calculations performed with the G3B3 compound method and treated with the CanTherm code [25]. This code is based on the atomization scheme for the computation of enthalpies of formation, and includes Bond Additivity Corrections (BAC) that are specific to the G3B3 method. The output of the code consists of enthalpy of formation, entropy, and constant pressure heat capacities at several temperatures, which were finally fitted into NASA polynomials. Thermochemistry for other species involved in the low- and high-temperature chemistry of the fuel were taken from Cai et al. [9] and are based on the group additivity method [26].

The nomenclature proposed by Cai et al. [9] was adopted and the carbon directly in the vicinity of the oxygen atom is called "a", while the following carbon is called "b" and so on. The radical resulting from the H-abstraction reaction on carbon "a" of DBE is therefore called "$C_4H_9OC_4H_8$-a" (Fig. 2).

4.1. High-temperature chemistry

Initiations by C–H bond breaking are written in the radical-radical recombination direction and the rate constant is set to 1×10^{14} s^{-1}. C–C bond breaking are considered by analogy with butanol. The rate constant for the unimolecular decomposition reaction DBE $\rightarrow$ 1-C_4H_8 + C_4H_9OH is a result from present calculations and is considered as pressure- dependent. In the study of Cai et al. [9], rate constants of H-abstraction reactions by hydroxyl radical were estimated by analogy with butanol and were increased by a factor of 1.5. No correction was applied in the present study and the rate constants used were those for butanol. The same type of analogy was used for H-abstraction reactions by O, C_2H_5, CH_3O, CH_3O_2 radicals and for the initiation reactions with O_2. Rate constants of the H-abstraction reactions by H atoms are those calculated by Al-Rashidi et al. [10] and by HO_2 radicals are those proposed by Mendes et al. [27] for ethers. Rate constants for the β-scissions of the fuel radicals were obtained from our calculations and H-abstraction, isomerization and β-scission reactions were added to consume heavy species produced by the decomposition of the fuel radicals. Rate constants for these reactions were estimated by analogy with DBE.

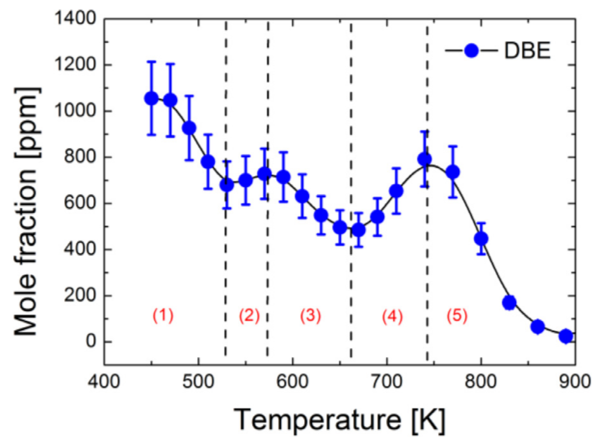

Fig. 3. Fuel mole fraction profile under fuel-rich conditions (φ = 2), at 10 atm and for τ = 700 ms. The black curve is a trendline and symbols are experimental data.

4.2. Low-temperature chemistry

Rate constants for addition reactions of the fuel radical on O_2 were taken from the work of Goldsmith et al. [28]. Rate constants for isomerization reactions of RO_2, cyclic ether formation, ketohydroperoxyde formation and decomposition of the RO_2 to alkenes and HO_2 were adapted from Villano et al. [29,30] as follows:

- The activation energy of the rate constants for the RO_2 isomerization to QOOH was reduced by 3 kcal mol^{-1} when the H atom to be transferred was located near the ether group. The original rate constant proposed by Villano et al. [29,30] was used for the other cases.
- The rate constants for the cyclic ether formation was adopted with no alterations.
- The activation energy of the formation of the ketohydroperoxydes proposed by the authors was reduced by 3 kcal mol^{-1} when the hydroperoxy group and the H to be transferred are located on the α position (near the ether group). This is an estimation to take into account the effect of the two oxygenated groups on the C – H BDE.
- The rate constant for the decomposition of RO_2 to alkenes and HO_2 were adopted without any alteration.

The rate constants used by Cai et al. [9] for the ketohydroperoxyde decomposition is $2.0\times10^{16}\times\exp(-35{,}000/RT)$ s^{-1} which is very high as compared to the rate constants recommended by Sahetchian et al. [31] ($4\times10^{15}\times\exp(-43{,}000/RT)$) and by Vasudevan et al. [32] ($2.5\times10^{15}\times\exp(-43{,}000/RT)$), usually used for this type of reaction. This correction was applied by the authors to improve their model's predictions in the low-temperature region. This type of reaction mainly depends on the bond dissociation energy of the O–OH bond, which is very weak: the bond breaks and the formed radical then decomposes fast. The ether group can reduce the bond dissociation energies of the groups located in its vicinity which could justify such a correction. In the case of the OOH group in α position, the rate constant used in the present mechanism is empirically set to $4.0\times10^{15}\times\exp(-37{,}000/RT)$ s^{-1} while the conventional rate constant of $4.0\times10^{15}\times\exp(-43{,}000/RT)$ s^{-1} is used for the other cases.

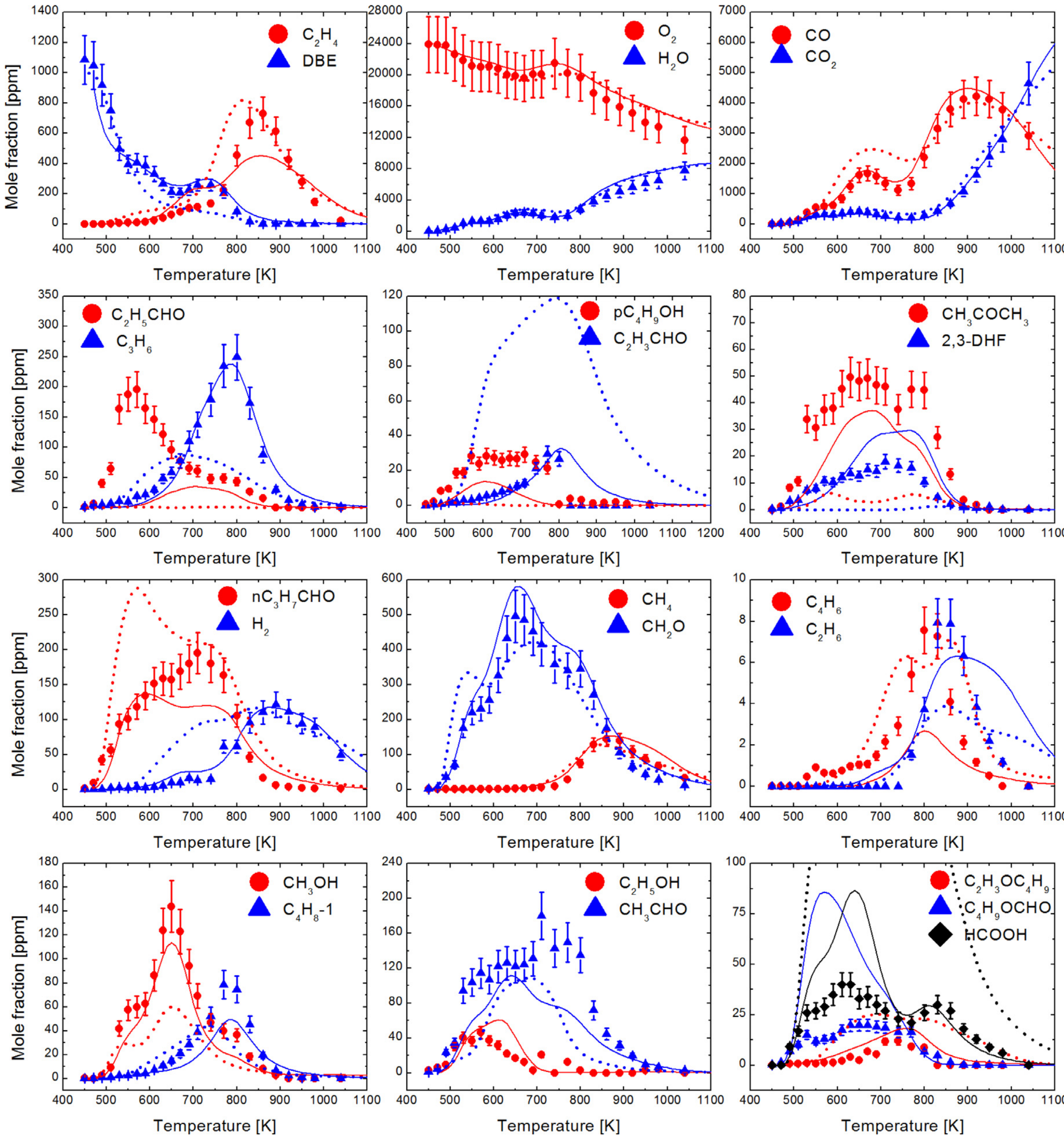

Fig. 4. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of DBE in a JSR at $\varphi = 0.5$, $p = 10$ atm, and $\tau = 0.7$ s. Solid lines: this work, dashed lines: Cai et al. [9].

5. Results

5.1. Oxidation at high-pressure

DBE is very reactive: it starts oxidizing at 460 K (Fig. 3); at a temperature much smaller than what was previously observed for other fuels under similar conditions [33–36]. Strong low-temperature chemistry is observed with the three mixtures at high-pressure. This chemistry is also entirely coupled with the high-temperature chemistry: once the threshold temperature is reached, the fuel consumption starts and its mole fraction never reaches its initial value when temperature rises (Fig. 3). The observed reactivity is also very peculiar as illustrated in Fig. 3. As a matter of fact, a "double cool-flame" is observed. For example, at $\varphi = 2$, the fuel consumption starts at c.a. 460 K and increases until 530 K (1). The consumption then slows down (2) and finally increases again at 580 K (3). Another negative temperature coefficient (NTC) is then observed at 660 K (4) and the reactivity increases again above 750 K (5). This unusual behavior is also observed under fuel-rich conditions (Fig. 6). The fuel consumption is much smoother under fuel-lean conditions (Fig. 4). A kinetic explanation to this phenomenon is provided in the next sections of this paper.

Figures 4–6 show the results obtained at 10 atm for a residence time of 700 ms and 1000 ppm of fuel under fuel-rich ($\varphi = 2$), stoichiometric, and fuel-lean ($\varphi = 0.5$) conditions. The main hydrocarbon intermediate is ethylene with mole fractions going up to 1200 ppm at $\varphi = 1$. Apart from CO, CO_2 and H_2O, the oxygenated species observed are mainly aldehydes and alcohols. 140 ppm of methanol, 40 ppm of ethanol, and 40 ppm of butanol were observed during the oxidation of the stoichiometric mixture. The main aldehyde observed is formaldehyde, followed by butanal, propanal, acetaldehyde and acrolein.

The simulation reasonably reproduces the fuel consumption but the first NTC observed between 530 and 570 K is underestimated

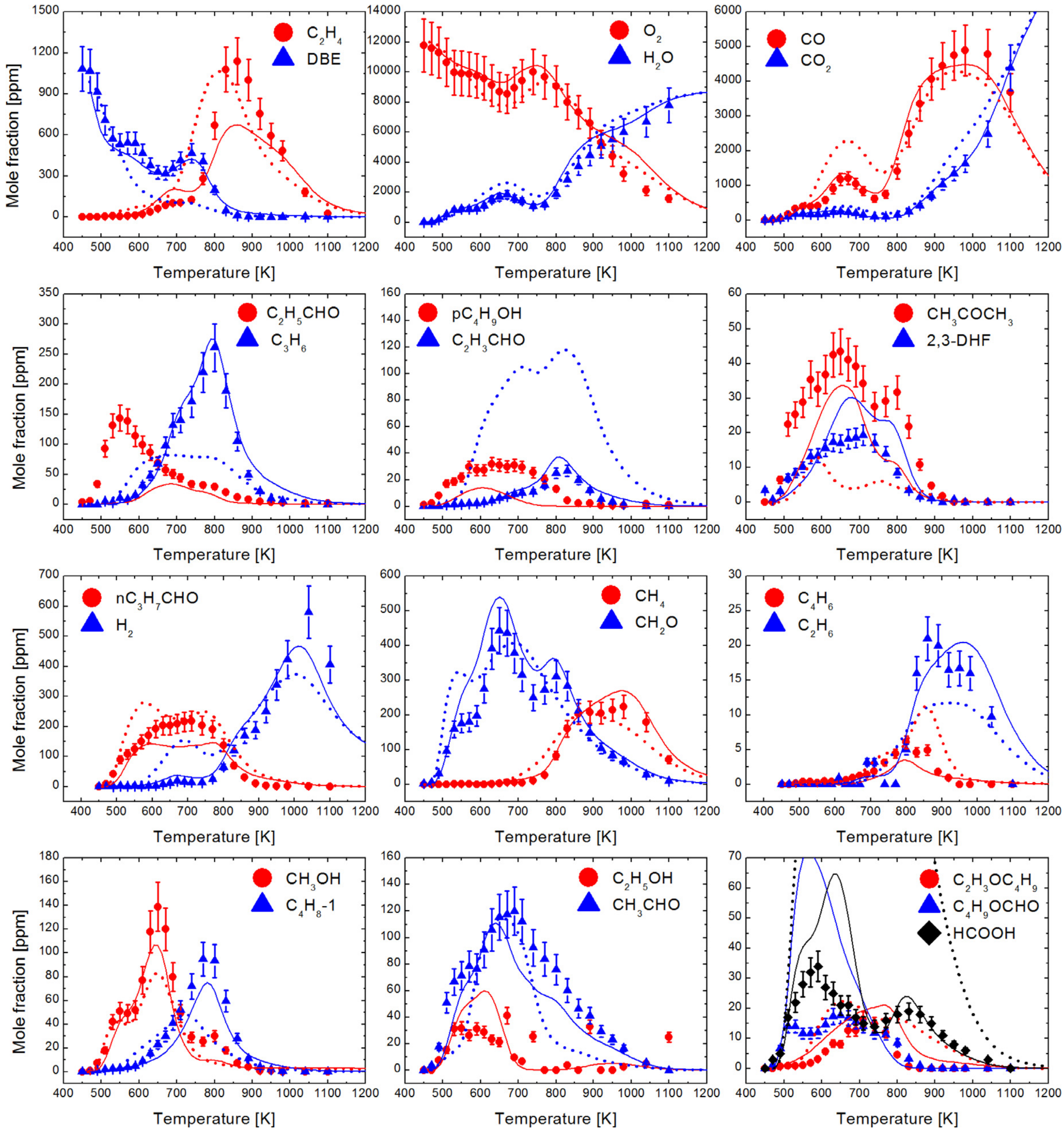

Fig. 5. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of DBE in a JSR at $\varphi = 1$, $p = 10$ atm, and $\tau = 0.7$ s. Solid lines: this work, dashed lines: Cai et al. [9].

by the model. The simulation with the mechanism proposed by Cai et al. [9] fails at reproducing these reactivity changes: the fuel is gradually consumed all over the temperature range, with no reactivity drop at all. Other profiles such as acrolein or formic acid are also overestimated using Cai's model [9]. Formic acid is also overpredicted by the mechanism developed in this study but to a lesser extent. This can be observed between 600 and 700 K for the fuel-lean and stoichiometric mixtures, and between 500 and 700 K under fuel-rich conditions, where the discrepancy between experimental data and modeling is more important. Formic acid is mainly produced from formaldehyde:

$$CH_2O + OH \leftrightarrows HOCH_2O \leftrightarrows HCOOH + H$$

The formaldehyde profile is slightly higher than the experimental data, and the difference between the simulation and the measurements at the peak mole fraction located at 650 K is c.a. 30 ppm at $\varphi = 1$. The overestimation of HCOOH mole fraction is of the same order of magnitude (~30 ppm) which could result from the overestimation of CH_2O. Several other species profiles show discrepancies between data and modeling for temperatures lower than 800 K: butyl-formate mole fractions are overestimated and propanal, 2, 3-dihydrofuran, acetone, and butanal mole fractions are underestimated. This shows that, despite the overall good performance of the present mechanism for the major species, there is still room for improvements for this low-temperature region that involves a very complex chemistry.

5.2. Qualitative pathway analysis for the low-temperature oxidation of DBE

The two NTC regions observed at 10 atm exhibit an interesting feature of DBE oxidation. The fuel mole fraction profile can be divided into five distinct temperature windows, as shown in Fig. 3.

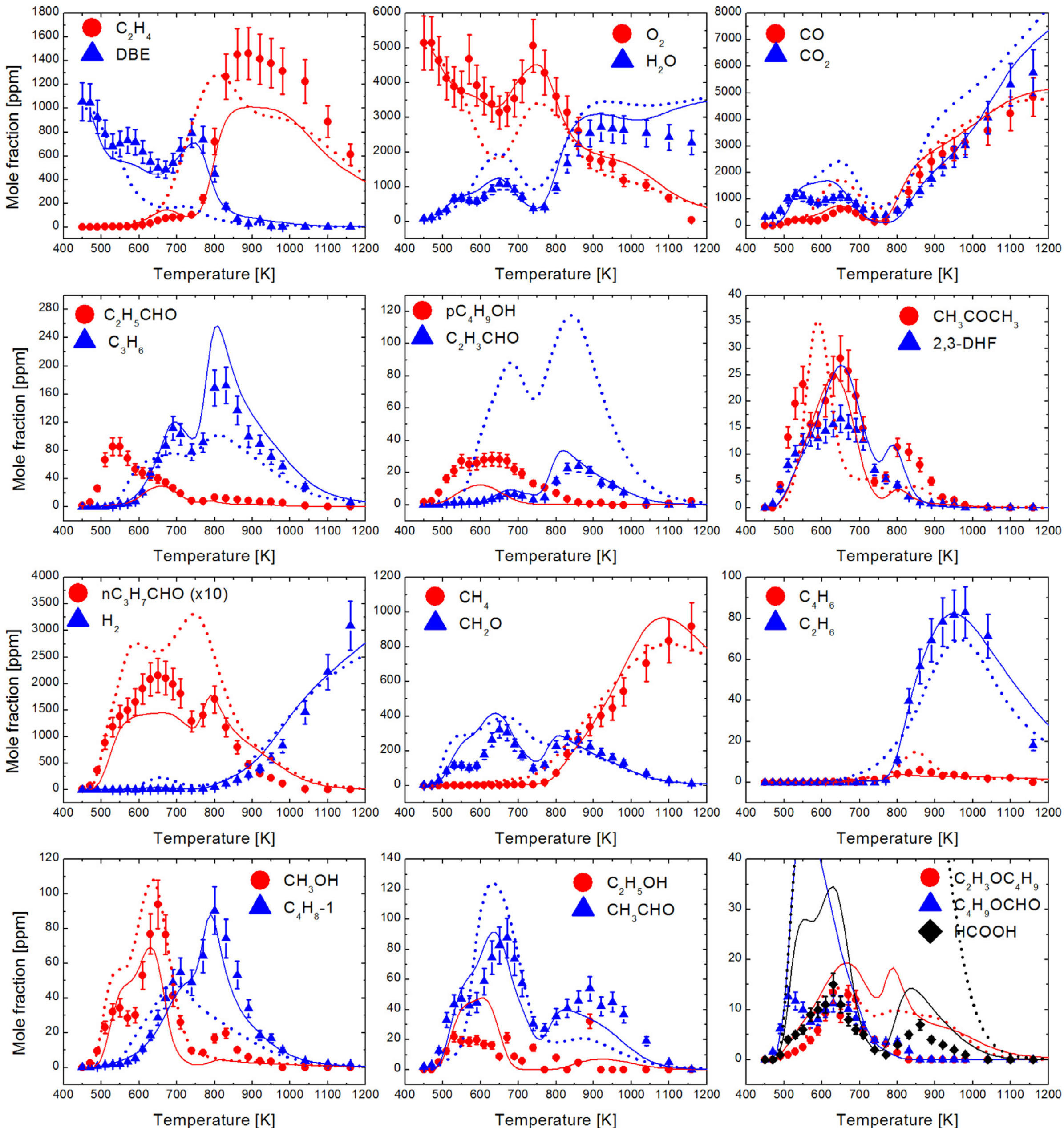

Fig. 6. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of DBE in a JSR at $\varphi = 2$, $p = 10$ atm, and $\tau = 0.7$ s. Solid lines: this work, dashed lines: Cai et al. [9].

Although some discrepancies are observed between the simulation and the experimental data in the 550–650 K temperature interval, a reaction pathway analysis was carried out with our model in an attempt to delineate the main chemical routes responsible for the unusual oxidation behavior of DBE. This analysis is illustrated by Fig. 7 and shows that DBE is mainly consumed by H-abstraction reactions producing the $C_4H_9OC_4H_8$-a radical. The different reaction pathways involved in the decomposition of this radical are therefore responsible for the reactivity switches observed on the fuel profile.

In region 1 ($T \leq 530$ K), the reactivity is triggered by a reaction sequence leading to the formation of a ketohydroperoxide (named “C4OC4keta-1”) and the hydroxyl radical. This ketohydroperoxyde is the most favorable as it involves H-transfer from the α position (neighboring the ether group) which exhibits the weakest C–H bond dissociation energy [9]. This is the main source of hydroxyl radicals in this region, as illustrated by the bottom right panel of Fig. 7(a). In this temperature range, the decomposition of C4OC4keta-1 is slow and its concentration profile rises with the temperature, as illustrated by the top right panel of Fig. 7. Its decomposition, however, starts as temperature increases (region 2) and leads to the formation of C_3H_7OOH which is produced through a low-temperature oxidation sequence of the propyl radical. The activation energy for the decomposition of C_3H_7OOH is higher than that of C4OC4keta-1 (42 vs. 37 kcal mol^{-1}) because of the ether group which lowers the O–OH bond dissociation energy for the latter one. This explains why the concentration profiles of these two intermediates start at different temperatures, as illustrated on Fig. 7 (top). The formation of C_3H_7OOH is more important than its decomposition in region 2 and its concentration rises while the reactivity stabilizes, producing the first NTC observed in temperature region 2.

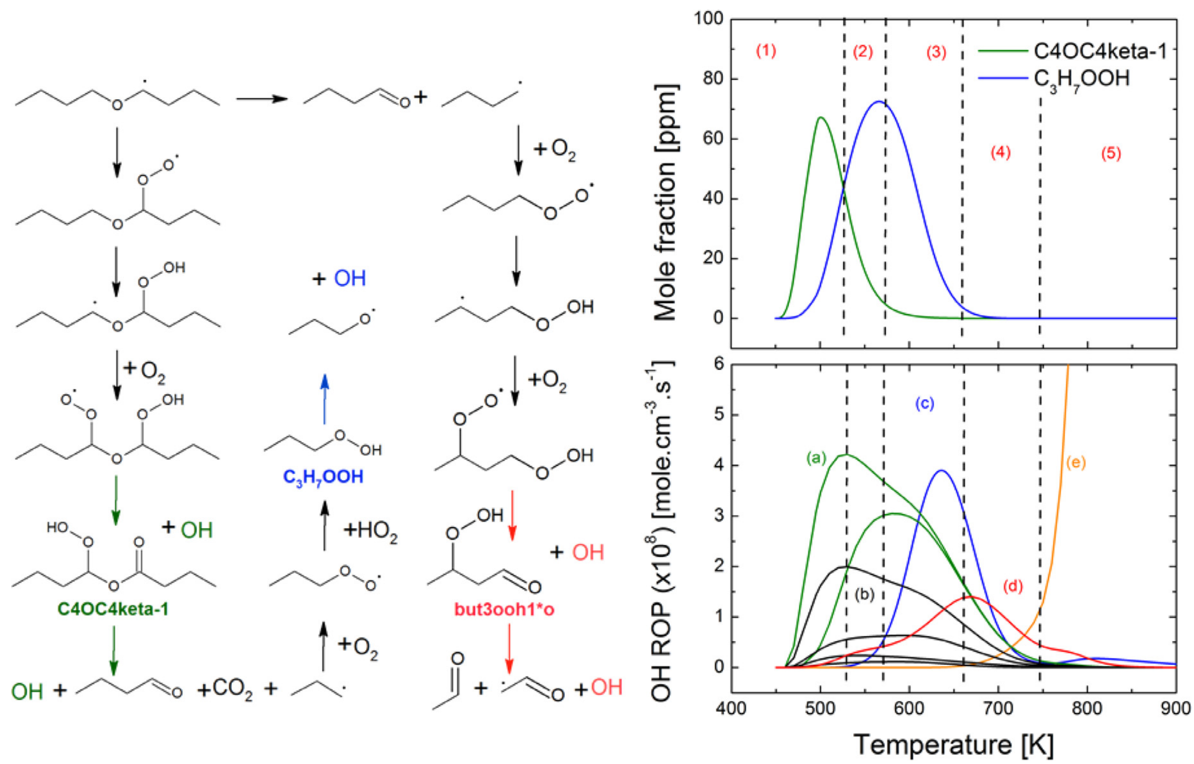

Fig. 7. Left: Main chemical reactions responsible for the different temperature regimes observed for the oxidation of DBE in a JSR obtained from the simulation at $\varphi = 2$. Top right: mole fraction profile of C4OC4keta-1 and C_3H_7OOH. Bottom right: contribution of various reactions to the OH formation: (a) formation and decomposition of C4OC4keta-1; (b) formation of other C8 ketohydroperoxydes; (c) decomposition of C_3H_7OOH; (d) decomposition of but3ooh1*o; (e) H_2O_2 decomposition.

The decomposition of C_3H_7COOH really starts in region 3, resulting in a drop of its concentration profile and a reactivity increase (Fig. 3) due to this new pathway for the formation of the hydroxyl radical. This decomposition becomes the main pathway forming OH in region 3; it is strong enough to counterbalance the reduced contribution of reactions involving C4OC4keta-1 on the hydroxyl radical production.

In the 660–740 K temperature range (region 4), the O_2 addition on the fuel radical $C_4H_9OC_4H_8$-a becomes less favorable, which inhibits the low-temperature chemistry and reduces the reactivity. This region 4 corresponds to the second negative temperature coefficient before the transition from low- to high-temperature chemistry. At temperatures higher than 740 K (region 5), the equilibrium of the O_2 addition is completely shifted towards the reactants and the β-scission reaction producing butanal and the but-1-yl radical, $C_4H_9OC_4H_8\text{-a} \rightleftharpoons nC_4H_9CHO + pC_4H_9$ becomes responsible for the consumption of $C_4H_9OC_4H_8$-a.

Aside from this chemistry directly related to the fuel, reactions of the butyl radical are also influencing the reactivity, as illustrated by the rate of production analysis for OH. This radical is mainly produced from the β-scission of the $C_4H_9OC_4H_8$-a fuel radical above 600 K. The formation of the butyl radical is slow at 600 K but gets faster as the temperature increases. This radical is then transformed by a sequence of low-temperature reactions illustrated by Fig. 7, which results in the formation of hydroxyl radicals. This reaction network takes place at 600–800 K and therefore increases the reactivity in region (3) and supports the transition from low- to high-temperature chemistry of DBE in region (4).

The low-temperature reactivity of the fuel is thus ruled by a complex combination of the low-temperature chemistry of 3 radicals ($C_4H_9OC_4H_8$-a, C_3H_7 and C_4H_9), which explains the complex behavior observed on the mole fraction profiles and makes the modeling task more difficult.

5.3. Oxidation at atmospheric pressure

Results obtained for experiments at atmospheric pressure are illustrated in Figs. 8–10. Low-temperature reactivity is observed at $\varphi = 1$ and $\varphi = 0.5$, but to a much lesser extent than at 10 atm. This reactivity can be observed for temperatures lower than 700 K while high-temperature reactivity can be observed above 950 K in all conditions investigated. No reactivity was observed at $\varphi = 2$ below 950 K and measurements performed for temperatures lower than 800 K were omitted from Fig. 10 for the sake of clarity. The reactivity issues already observed with the mechanism of Cai et al. [9] at higher pressure are highlighted here since the simulation with their mechanism at $\varphi = 2$ yields a fuel consumption starting at 800 K in the high-temperature region. The chemical reactions responsible for this high-temperature discrepancy are the H-abstraction reactions from DBE by the HO_2 radical. The rate constants used by Cai et al. [9] for these reactions have been estimated by analogy with dimethyl-ether. Their total rate constant is 200 times faster than that calculated by Mendes et al. [27] at 800 K and 100 times faster at 1100 K. Replacing these estimated rate constants by those proposed by Mendes et al. [27] for methyl-butyl-ether results in improved predictions of the fuel profile as well as mole fraction profiles of products and intermediate species, which were otherwise overpredicted. Another factor contributing to the overpredictions with the literature model is the unimolecular decomposition of the ketohydroperoxydes. The rate constants used by Cai et al. [9] for these reactions are very fast, as explained previously, and contribute to the reactivity increase in the low-temperature region.

The mechanism developed herein yields very satisfactory simulation results for all mixtures investigated in this work. Main discrepancies at 1 atm between modeling and experimental data concern butanol which is underestimated in both temperature regions

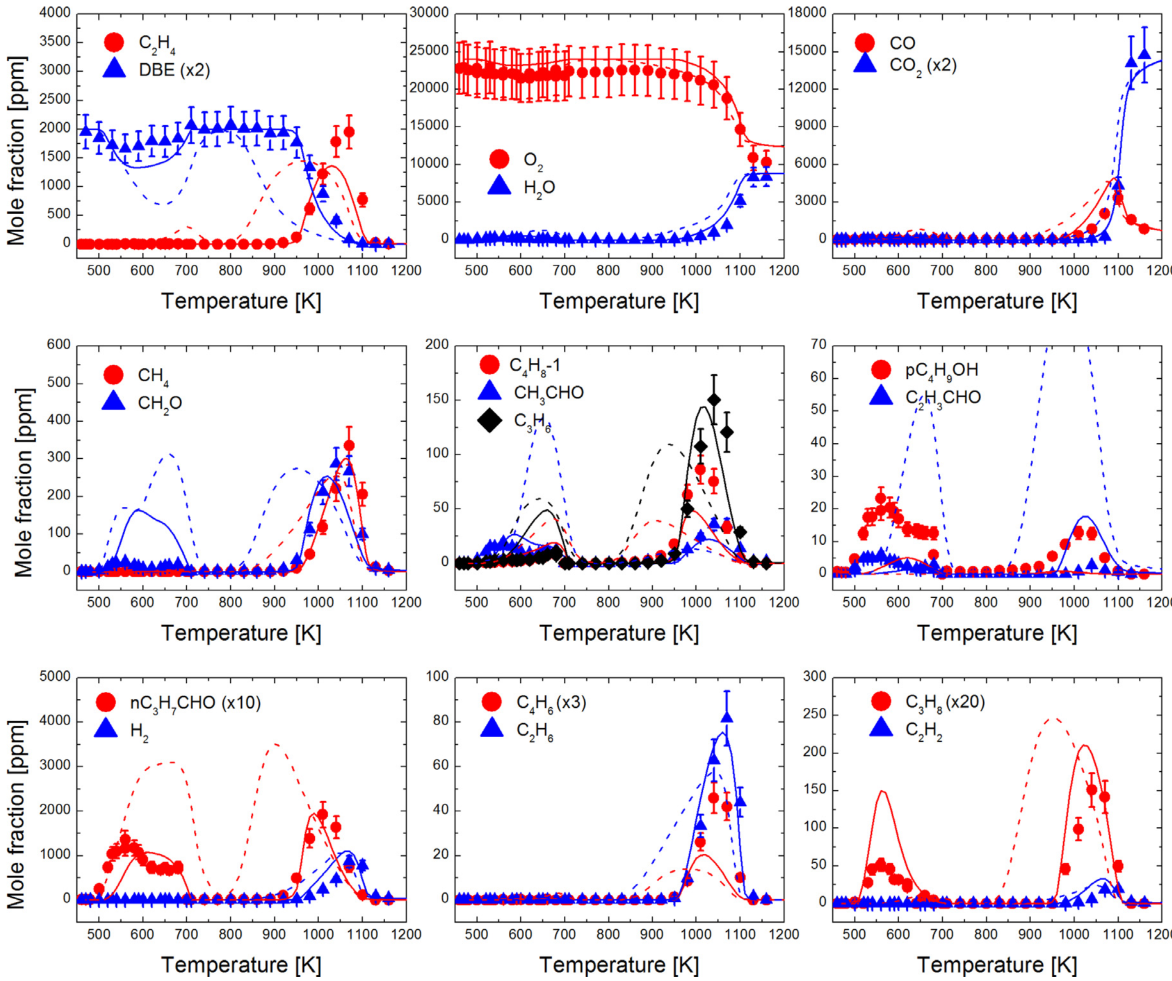

Fig. 8. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of DBE in a JSR at $\varphi = 0.5$, p = 1 atm, and $\tau = 0.07$ s. Solid lines: this work, dashed lines: Cai et al. [9].

and formaldehyde, which is overestimated in the low-temperature region. Issues in this region echo with what was already observed at 10 atm and show that improvements are still necessary on the low-temperature subset of DBE. The formation of CH_2O indeed stems from the decomposition of alkoxy radicals (mainly C_4H_9O and C_3H_7O) that are produced by the decomposition of ketohydroperoxydes.

The reaction responsible for the formation of butanol in the high-temperature region is the unimolecular decomposition of the fuel ($C_4H_9OC_4H_9 \rightleftharpoons C_4H_8\text{-}1 + C_4H_9OH$). The pressure-dependent rate constant used in this work for this reaction was calculated by the method described in Section 3 of this paper. This rate constant is slightly higher than what was computed by Al Rashidi [10] for the same reaction (by a factor of 3 at 1000 K) but is not high enough to reproduce the experimental data. This suggests that the rate constant computed for this reaction with the G3B3 and G4 methods are too low. The rate constant computed in this study would have to be multiplied by a factor of 15 to make the simulation reach the peak butanol mole fraction measured for the stoichiometric mixture. A missing pathway leading to the formation of butanol could also be considered to explain the discrepancy observed between the data and the modeling.

5.4. Rate of production analysis in the high-temperature regime

The atmospheric pressure dataset enables the validation of the high-temperature sub-mechanism because low- and high-temperature reactivities are clearly decoupled, which was not the case at 10 atm. As previously discussed the simulation with our mechanism yields good performances and can therefore be used to get more insights into the high temperature oxidation mechanism of DBE. A rate of production analysis (ROP) was performed for the stoichiometric mixture at 1000 K, which roughly corresponds to 50% of the fuel consumption (Fig. 11).

This analysis shows that DBE is mainly consumed (51%) to produce the radical $C_4H_9OC_4H_8$-a. This radical then decomposes to butanal and the but-1-yl radical (82%) while the contribution of the decomposition of radical “a” leading to the formation of ethenyloxibutane and ethyl radical is more modest (18%). The second most important radical is the radical $C_4H_9OC_4H_8$-c which either decomposes to propene and $C_4H_9OCH_2$ or isomerizes to $C_4H_9OC_4H_8$-a. 14% of the fuel consumption also occurs through H-abstraction reactions leading to the formation of the radical $C_4H_9OC_4H_8$-d. The fate of this radical is similar to that of $C_4H_9OC_4H_8$-c. 73% of its conversion represents isomerization to radical $C_4H_9OC_4H_8$-a while its decomposition to ethylene and $C_4H_9OC_2H_4$ has a contribution of 24%. H-abstraction reactions on DBE to produce the radical $C_4H_9OC_4H_8$-b are the least favorable ones and are only responsible for 10% of the fuel consumption. This radical can undergo two different β-scission reactions to form either 1-butene and butoxy radical (85%) or methyl radical and $C_4H_9OC_3H_5$ (15%). This ROP analysis highlights the central role of the $C_4H_9OC_4H_8$-a radical, as it is the major product of H-abstraction reactions on DBE as well as the main product of the

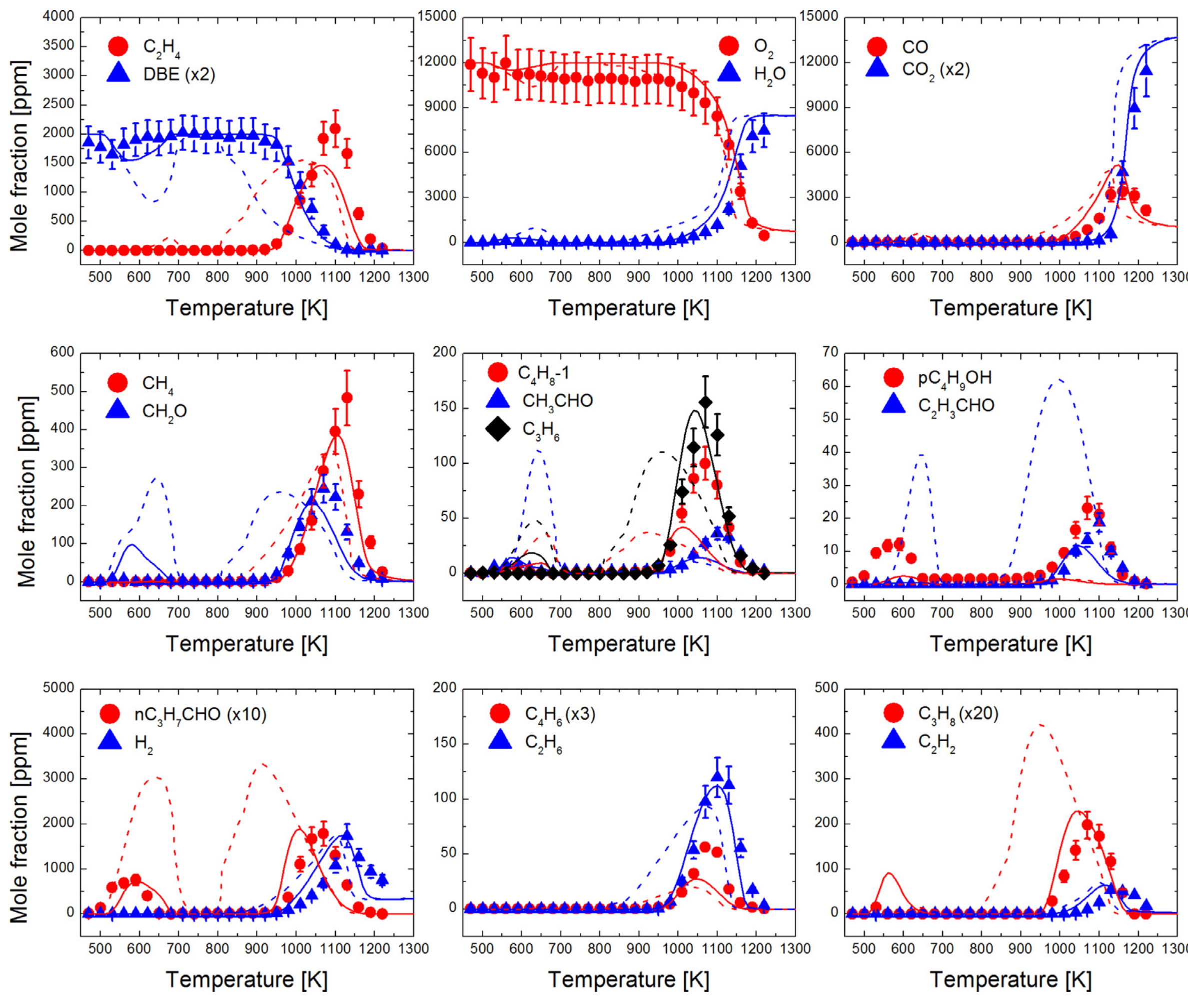

Fig. 9. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of DBE in a JSR at $\varphi = 1$, p = 1 atm, and $\tau = 0.07$ s. Solid lines: this work, dashed lines: Cai et al. [9].

radicals $C_4H_9OC_4H_8$-c and $C_4H_9OC_4H_8$-d through isomerizations. This explains the relatively high mole fraction of butanal observed (~200 ppm for all mixtures) and of ethylene, which is mainly produced through the decomposition of the butyl radical.

6. Further validation

Ignition delay times measured by Guan et al. [11] were used for further validation of our detailed kinetic reaction mechanism. Figure 12 shows a comparison between these experimental data (symbols) and simulations using our model and that of Cai et al. [9] at atmospheric pressure. Results obtained at higher pressures (2 and 4 atm) are available in the supplementary material. The two simulations yield very similar results and are in excellent agreement with the data of Guan et al. [11] under the different experimental conditions.

The simulation of the low temperature ignition delay times of Cai et al. [9] is illustrated by Fig. 13. The authors observed that the ignition delay time was not sensitive to the equivalence ratio in their conditions, so the figure shows the experimental data obtained for the three investigated mixtures (φ=0.5, 0.7 and 1) together for more clarity. The simulation with the mechanism of Cai et al. [9] produces slightly shorter ignition delay times than our mechanism in the coldest part of the temperature range, but the agreement is very good, with a maximum deviation of roughly 25% at 490 K.

Cai et al. [9] also measured laminar burning velocities of a DBE/air flames at 373 K and 1 atm and used them to test their mechanism. Figure 14 shows these data as well as simulations obtained with the mechanism developed in this study and that of Cai et al. [9]. The two simulations are very similar and both tend to overestimate the laminar burning velocity under fuel-rich conditions. Performances are very good for equivalence ratios from 0.7 to 1.2. The maximum deviation between experiments and simulations can be observed at $\varphi = 1.5$, where the simulation with the mechanism developed herein is higher by 3 cm s^{-1} from that calculated with the mechanism of Cai et al. [9]. The performances of both mechanisms are still very reasonable, as the uncertainty on the measurements is larger under fuel-rich conditions (±3.5 cm s^{-1} at $\varphi = 1.5$) than for fuel-lean mixtures (±1.5 cm s^{-1} at $\varphi = 0.7$).

The performances of our mechanism are similar to those obtained by the mechanism proposed by Cai et al. [9] for laminar burning velocities and high temperature ignition delay times. This must be added to the fact that simulations of our experimental JSR data with our mechanism yields improved performances as compared to similar simulations with the model of Cai et al. [9].

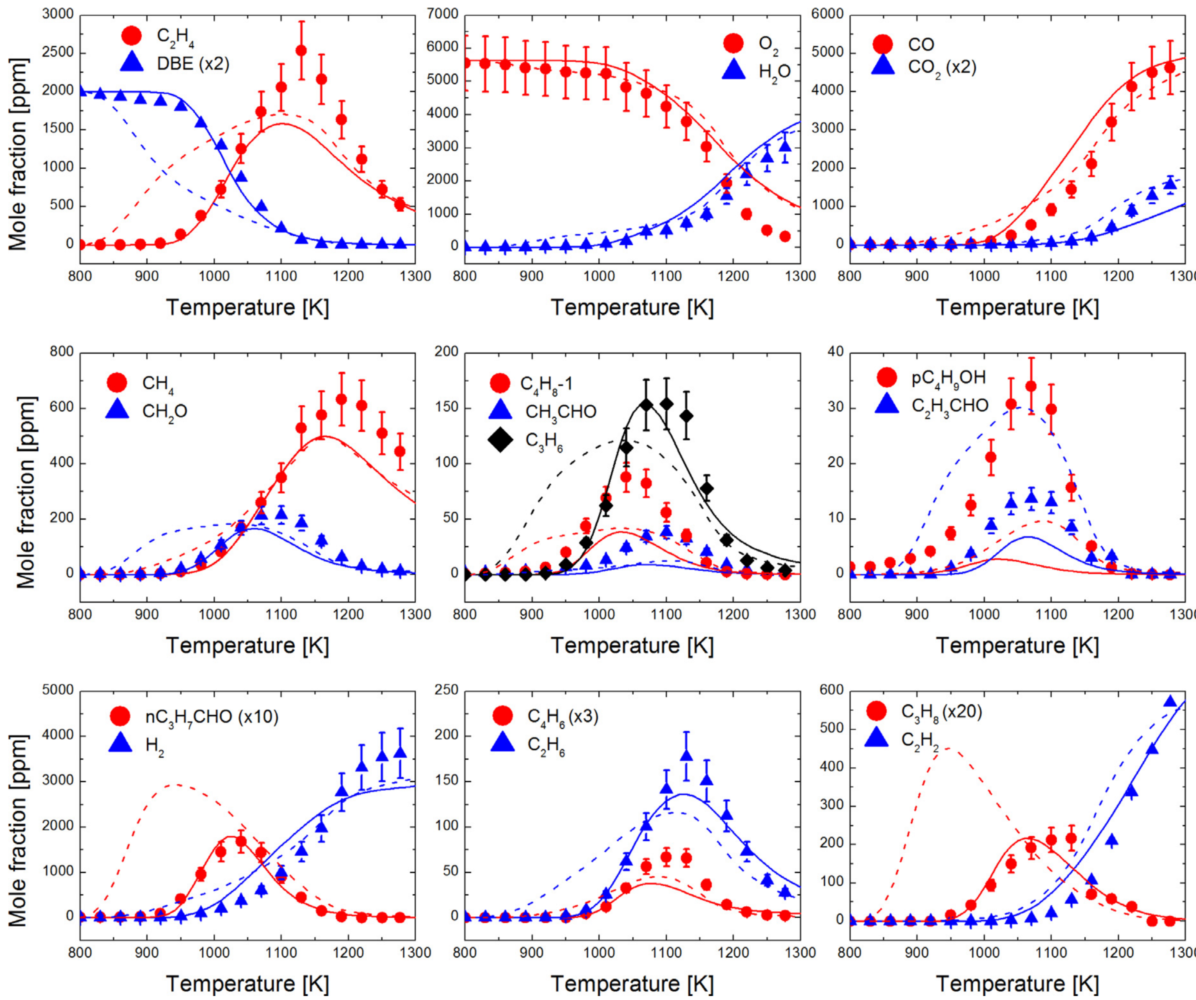

Fig. 10. Experimental (symbols) and computed (lines) mole fraction profiles obtained from the oxidation of DBE in a JSR at $\varphi = 2$, p = 1 atm, and $\tau = 0.07$ s. Solid lines: this work, dashed lines: Cai et al. [9].

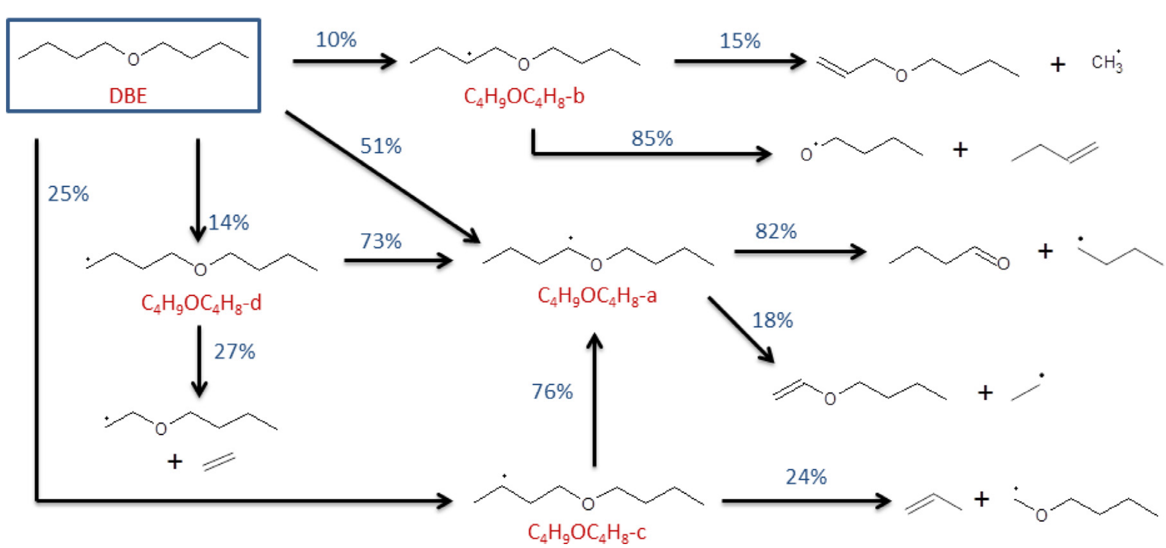

Fig. 11. Rate of production analysis for the oxidation of a stoichiometric mixture in a JSR at 1atm and 1000 K.

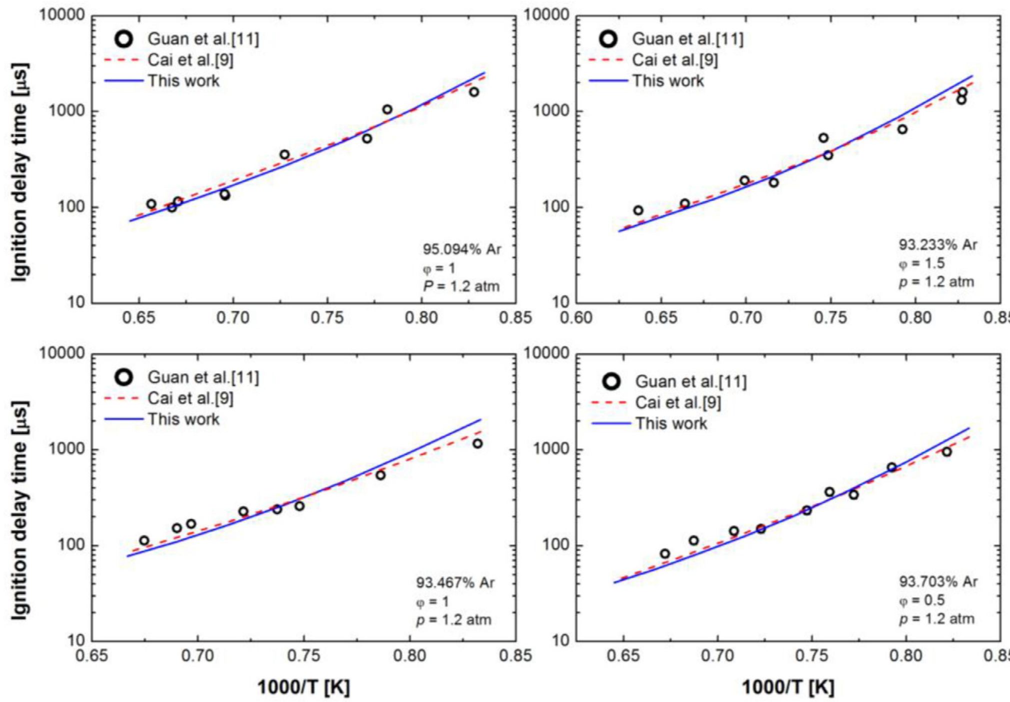

Fig. 12. Experimental ignition delay times of DBE/O_2/Ar mixtures in a shock-tube [11] (symbols) and simulations with the present mechanism (solid blue line) and the mechanism of Cai et al. [9] (dashed red lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

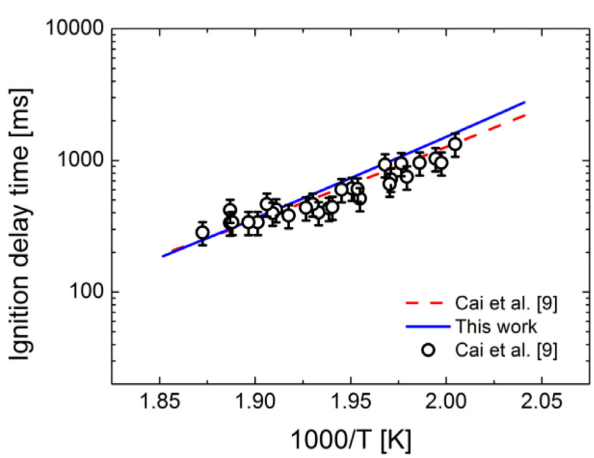

Fig. 13. Experimental ignition delay times of DBE/air in a laminar flow reactor (symbols) [9] and simulations with the present mechanism (solid blue line) and the mechanism of Cai et al. [9] (dashed red lines). Symbols represent all equivalence ratios data. Simulation performed at $\varphi = 1$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

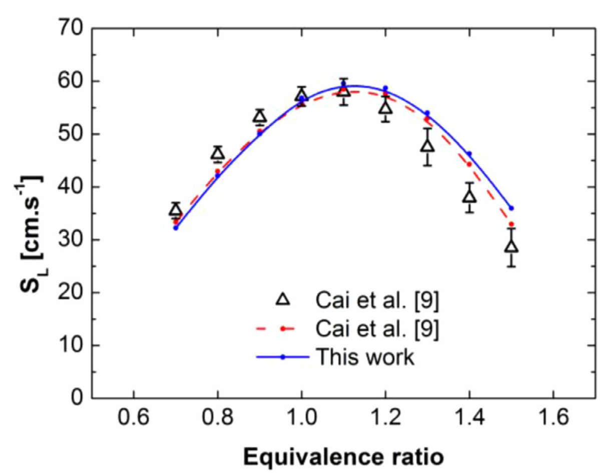

Fig. 14. Experimental laminar burning velocity of DBE/air mixtures at atmospheric pressure and $T_{\mathrm{unburnt}} = 373$ K (symbols), and simulations with the present mechanism (solid blue line) and the mechanism of Cai et al. [9] (dashed red lines). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

7. Conclusion

The oxidation of DBE was studied in a jet-stirred reactor under fuel-lean, stoichiometric, and fuel-rich conditions at 1 and 10 atm. This potential biofuel exhibited a strong and complex low-temperature oxidation chemistry with two negative temperature coefficients at high pressure. A detailed kinetic mechanism was developed and validated based on these data. The best model performances are obtained at high temperature while improvements are still possible in the low-temperature region. The fuel oxidizes mainly through the formation of the $C_4H_9OC_4H_8$-a radical which is the starting point of the reactivity. At high temperature, this radical mainly decomposes into the but-1-yl radical and butanal which is a major oxygenated intermediate of DBE oxidation. In the low-temperature region, the $C_4H_9OC_4H_8$-a radical chemistry is responsible for the reactivity oscillations observed during the experiments. It is consumed by a reaction network producing a ketohydroperoxyde, which in turn decomposes and yields the prop-1-yl radical. This latter radical is consumed through the production of a hydroperoxyde which finally decomposes to C_3H_7O and the hydroxyl radical. The butyl radical, which is produced from radical $C_4H_9OC_4H_8$-a and from the decomposition of heavier low-temperature intermediates, is also a key species for the reactivity through its low-temperature reactions. The 2 NTCs regions observed during DBE oxidation therefore result from the combination of the low-temperature chemistry of the $C_4H_9OC_4H_8$-a, C_4H_9 and C_3H_7 radicals.

The mechanism developed here yields more accurate simulations on our JSR data than a model available in the literature. Further model validations were also accomplished by successfully modeling ignition delay times and laminar burning velocities previously reported. The discrepancies between data and model on some concentration profiles at low-temperature highlights the need for more insight regarding the complex chemistry involved in the low-temperature oxidation of DBE. Investigations regarding the effect of the ether functional group on the decomposition of low-temperature intermediates could be an interesting starting point for further works.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2017.06.019.

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